

Linux Basic Commands

Command: whoami

- Description: Displays the current user account.
- Example: Running `whoami` in the terminal will show your username, like `john_doe`.

```
nbkx786@techies:~$  
nbkx786@techies:~$ whoami  
nbkx786  
nbkx786@techies:~$  
nbkx786@techies:~$
```

Command: ls

- Description: Lists files and directories in the current directory.
- Example: `ls` will display files and folders in your current directory, and `ls -l` provides detailed information.

```
nbkx786@techies:~$  
nbkx786@techies:~$ ls  
Desktop  Downloads  Music  Pictures  Public  Templates  Videos  d1  d2  m1  snap  
nbkx786@techies:~$  
nbkx786@techies:~$
```

Command: ip a

- Description: Lists your IP address and network interfaces.
- Example: `ip a` show all network interfaces and their IP addresses.

```
nbkx786@techies:~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default  
    link/ether 46:2f:2c:44:3b:ce brd ff:ff:ff:ff:ff:ff  
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0  
        valid_lft forever preferred_lft forever  
3: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:09:11:38 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.225.129/24 brd 192.168.225.255 scope global dynamic noprefixroute ens33  
        valid_lft 1748sec preferred_lft 1748sec  
    inet6 fe80::20c:29ff:fe09:1138/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
nbkx786@techies:~$
```

Command: cat

- Description: Reads the contents of text files.
- Example: `cat myfile.txt` will display the content of `myfile.txt`.

```
nbkx786@techies:~$ cat myfile.txt  
Linux basic command practice !!!  
Hello World !!!  
nbkx786@techies:~$  
nbkx786@techies:~$
```

Command: passwd

- Description: Used to change the password of the user account.
- Example: Typing `passwd` and following the prompts will allow you to change your password.

```
nbkx786@techies:~$ passwd  
Changing password for nbkx786.  
Current password: █
```

Command: touch

- Description: Creates empty files, mainly useful to test write ability.
- Example: `touch newfile.txt` creates an empty file named `newfile.txt`.

```
nbkx786@techies:~$  
nbkx786@techies:~$ touch newfile.txt  
nbkx786@techies:~$  
nbkx786@techies:~$  
nbkx786@techies:~$ ls newfile.txt -ltr  
-rw-rw-r-- 1 nbkx786 nbkx786 0 Aug  6 14:03 newfile.txt  
nbkx786@techies:~$
```

Command: pwd

- Description: Stands for Print Working Directory; shows which directory you are in.
- Example: Running `pwd` might return `/home/john_doe`.

```
nbkx786@techies:~$  
nbkx786@techies:~$ pwd  
/home/nbkx786  
nbkx786@techies:~$
```

Command: man

- Description: Accesses the system documentation for commands.
- Example: `man ls` will show the manual for the `ls` command.

```
nbkx786@techies: ~ — man ls  
LS(1) General Commands Manual LS(1)  
  
NAME  
ls - List directory contents. Ignore files and directories starting with a '.' by default  
  
SYNOPSIS  
ls [--help] [--format] [-C ] [-l|--long] [-x ] [-T|--tabsize] [-m ] [--zero] [-D|--dired] [--hyperlink] [-1 ] [-o ] [-g ] [-n|--nu-  
meric-uid-gid] [--quoting-style] [-N|--literal] [-b|--escape] [-Q|--quote-name] [-q|--hide-control-chars] [--show-control-chars] [--time] [-c  
] [-u ] [--hide] [-I|--ignore] [-B|--ignore-backups] [--sort] [-S ] [-t ] [-v ] [-X ] [-U ] [-L|--dereference] [--dereference-com-  
mand-line-symlink-to-dir] [-H|--dereference-command-line] [-G|--no-group] [--author] [-a|--all] [-A|--almost-all] [-F ] [-d|--directory]  
[-h|--human-readable] [-k|--kibibytes] [--si] [--block-size] [-i|--inode] [-r|--reverse] [-R|--recursive] [-w|--width] [-s|--size] [--color]  
[--indicator-style] [-F|--classify] [--file-type] [-p ] [--time-style] [--full-time] [-Z|--context] [--group-directories-first] [-V|--version]  
[paths]  
  
DESCRIPTION  
List directory contents. Ignore files and directories starting with a '.' by default  
  
OPTIONS  
--help Print help information.  
  
--format  
Set the display format.  
  
-C Display the files in columns.  
  
-l, --long Display detailed information.  
  
-x List entries in rows instead of in columns.  
  
-T, --tabsize <COLS>  
Assume tab stops at each COLS instead of 8  
  
-m List entries separated by commas.  
  
--zero List entries separated by ASCII NUL characters.  
  
-D, --dired  
Manual page ls(1) line 1 (press h for help or q to quit)
```

Command: apropos

- Description: Searches commands related to a specific task based on description (equivalent to `man -k`).
- Example: `apropos "list files"` might return commands related to listing files.

Command: pinfo

- Description: Provides additional information, similar to `man`.
- Example: `pinfo ls` gives detailed information about the `ls` command.

```
root@techies: /home/nbkx786 — sudo bash  
File: coreutils.info, Node: ls invocation, Next: dir invocation, Up: Directory listing  
  
10.1 'ls': List directory contents  
=====
```

The 'ls' program lists information about files (of any type, including directories). Options and file arguments can be intermixed arbitrarily, as usual. Later options override earlier options that are incompatible.

For non-option command-line arguments that are directories, by default 'ls' lists the contents of directories, not recursively, and omitting files with names beginning with '.'. For other non-option arguments, by default 'ls' lists just the file name. If no non-option argument is specified, 'ls' operates on the current directory, acting as if it had been invoked with a single argument of '.'.

By default, 'ls' lists each directory's contents alphabetically, according to the locale settings in effect.(1) If standard output is a terminal, the output is in columns (sorted vertically) and control characters are output as question marks; otherwise, the output is listed one per line and control characters are output as-is.

Because 'ls' is such a fundamental program, it has accumulated many options over the years. They are described in the subsections below; within each section, options are listed alphabetically (ignoring case). The division of options into the subsections is not absolute, since some options affect more than one aspect of 'ls's operation.

Exit status:

- 0 success
- 1 minor problems (e.g., failure to access a file or directory not specified as a command line argument. This happens when listing a directory in which entries are actively being removed or renamed.)
- 2 serious trouble (e.g., memory exhausted, invalid option, failure to access a file or directory specified as a command line argument or a directory loop)

```
Showing line 34/54, 67%
```

Command: `tldr`

- Description: Shows command line examples, a simplified version of `man`.
- Example: `tldr passwd` provides quick examples of how to use the `passwd`

```
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# tldr passwd

Change a user's password.
See also: `chpasswd`.
More information: <https://manned.org/passwd>.

Change the password of the current user interactively:

    passwd

Change the password of a specific user:

    sudo passwd username

Get the current status of the user:

    passwd \[-S|--status\]

Make the password of the account blank (it will set the named account passwordless):

    passwd \[-d|--delete\]

Set password programmatically (ideal for install scripts):

    yes password | passwd

root@techies:/home/nbkx786#
```

Essential File Management Tools Commands

Command: `ls`

- Example: `ls /home/user/Documents`
- Description: Lists files and directories within a specified path. This command is used to view the contents of directories.

```
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786# ls  
Desktop Documents Downloads Music Pictures Public Templates Videos d1 d2 m1 myfile.txt newfile.txt snap  
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786#
```

Command: `cp`

- Example: `cp source.txt destination.txt`
- Description: Copies files from one location to another. This command is used to make duplicates of files.

```
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786# cp source.txt destination.txt  
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786# ls *.txt -la  
-rw-r--r-- 1 root root 0 Aug 6 14:22 destination.txt  
-rw-rw-r-- 1 nbkx786 nbkx786 49 Aug 6 14:00 myfile.txt  
-rw-rw-r-- 1 nbkx786 nbkx786 0 Aug 6 14:03 newfile.txt  
-rw-r--r-- 1 root root 0 Aug 6 14:21 source.txt  
root@techies:/home/nbkx786#
```

Command: `mkdir`

- Example: `mkdir new_directory`
- Description: Creates a new directory. This command is used when you need to make a new folder.

```
root@techies:/home/nbkx786# mkdir new_directory  
root@techies:/home/nbkx786# ls -la new_directory/  
total 8  
drwxr-xr-x 2 root root 4096 Aug 6 14:24 .  
drwxr-x-- 19 nbkx786 nbkx786 4096 Aug 6 14:24 ..  
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786#
```

Command: `rmdir`

- Example: `rmdir empty_directory`
- Description: Removes a directory, but only if it's empty. This command is used for deleting empty folders.

```
root@techies:/home/nbkx786# rmdir new_directory/  
root@techies:/home/nbkx786#
```

Command: `mv`

- Example: `mv oldname.txt newname.txt`
- Description: Moves or renames files and directories. This command can change the location or the name of a file.

```

root@techies:/home/nbkx786# cat myfile.txt
Linux basic command practice !!!
Hello World !!!
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# mv myfile.txt new_file.txt
root@techies:/home/nbkx786# ls new_file.txt
new_file.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# cat new_file.txt
Linux basic command practice !!!
Hello World !!!
root@techies:/home/nbkx786#

```

Command: **rm**

- Example: `rm filename.txt`
- Description: Removes files or directories. This command deletes files or directories.
- Note: When combined with options `-r` (recursive) and `-f` (force), as in `rm -rf directory_name`, it becomes a powerful command that can delete directories and their contents without asking for confirmation.

```

root@techies:/home/nbkx786# ls *.txt
destination.txt  new_file.txt  newfile.txt  source.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# rm destination.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# ls *.txt
new_file.txt  newfile.txt  source.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#

```

```

root@techies:/home/nbkx786# ls F1/ -ltr
total 4
-rw-r--r-- 1 root root  0 Aug  6 14:35 source.txt
-rw-r--r-- 1 root root 49 Aug  6 14:35 new_file.txt
-rw-r--r-- 1 root root  0 Aug  6 14:35 newfile.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# rm -rf F1/
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# ls F1/ -ltr
ls: cannot access 'F1/': No such file or directory
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#

```

Advanced File Management Tools Commands

ln: Used for creating hard and symbolic links.

- Example: `ln -s source_file symbolic_link` creates a symbolic link to `source_file`.
- Description: Links a file or directory to another location.

```
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786# ln -s source.txt source_link  
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786#
```

```
-rw-r--r-- 1 root root 0 Aug 6 14:21 source.txt  
lrwxrwxrwx 1 root root 10 Aug 6 14:45 source_link -> source.txt  
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786# echo "Linux Practice" > source.txt  
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786# cat source_link  
Linux Practice  
root@techies:/home/nbkx786#
```

```
lrwxrwxrwx 1 root root 10 Aug 6 14:50 f3_dir -> F1/F2/F3  
lrwxrwxrwx 1 root root 19 Aug 6 14:55 f3file_link -> F1/F2/F3/F3File.txt  
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786# cat f3file_link  
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786# echo "Linux Basics practice" > f3file_link  
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786# cat f3file_link  
Linux Basics practice  
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786#
```

find: A utility to find files based on various properties.

- Example: `find /home -name "*.txt"` searches for text files in the `/home` directory.
- Description: Searches for files and directories in a given path.

```
lrwxrwxrwx 1 root root 10 Aug 6 14:50 f3_dir -> F1/F2/F3  
lrwxrwxrwx 1 root root 19 Aug 6 14:55 f3file_link -> F1/F2/F3/F3File.txt  
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786#  
root@techies:/home/nbkx786# find F1 -name "*.txt"  
F1/F2/F3/F3File.txt  
root@techies:/home/nbkx786#
```

locate: Used for quickly searching the database of system's contents.

- Example: `locate filename` finds the location of `filename`.
- Description: Fast file searching, requires running `updatedb` for an updated database.

```

root@techies:/home/nbkx786# locate F3File.txt
/home/nbkx786/F1/F2/F3/F3File.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#

```

which: Finds the location of executables in the PATH.

- Example: `which python` shows the path of the Python executable.
- Description: Useful for finding executable files in system's PATH.

```

root@techies:/home/nbkx786#
root@techies:/home/nbkx786# which python3
/usr/bin/python3
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#

```

tar: Archives multiple files into a single file.

- Example: `tar -cvf archive.tar /folder` creates an archive of a folder.
- Description: Combines multiple files into a tarball.

```

root@techies:/home/nbkx786# ls -ltr F1/F2/F3/
total 16
-rw-r--r-- 1 root root 22 Aug  6 14:56 F3File.txt
-rw-r--r-- 1 root root 49 Aug  6 15:31 new_file.txt
-rw-r--r-- 1 root root 51 Aug  6 15:31 newfile.txt
-rw-r--r-- 1 root root 15 Aug  6 15:31 source.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# tar -cvf archive.tar F1/F2/F3/
F1/F2/F3/
F1/F2/F3/newfile.txt
F1/F2/F3/new_file.txt
F1/F2/F3/source.txt
F1/F2/F3/F3File.txt
root@techies:/home/nbkx786# ls -ltr *.tar
-rw-r--r-- 1 root root 10240 Aug  6 15:32 archive.tar
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#

```

gzip: Compresses files.

- Example: `gzip file.txt` compresses `file.txt`.
- Description: Reduces the size of files using compression.

```

root@techies:/home/nbkx786#
root@techies:/home/nbkx786# ls -ltr F1/F2/F3/
total 16
-rw-r--r-- 1 root root 22 Aug  6 14:56 F3File.txt
-rw-r--r-- 1 root root 49 Aug  6 15:31 new_file.txt
-rw-r--r-- 1 root root 51 Aug  6 15:31 newfile.txt
-rw-r--r-- 1 root root 15 Aug  6 15:31 source.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# gzip F1/F2/F3/newfile.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# ls -ltr F1/F2/F3/
total 16
-rw-r--r-- 1 root root 22 Aug  6 14:56 F3File.txt
-rw-r--r-- 1 root root 49 Aug  6 15:31 new_file.txt
-rw-r--r-- 1 root root 76 Aug  6 15:31 newfile.txt.gz
-rw-r--r-- 1 root root 15 Aug  6 15:31 source.txt
root@techies:/home/nbkx786#

```

bzip2: A file compression utility.

- Example: `bzip2 file.txt` compresses `file.txt`.
- Description: Offers better compression ratios than `gzip` at the cost of speed.

```

root@techies:/home/nbkx786# ls -ltr F1/F2/F3/
total 16
-rw-r--r-- 1 root root 22 Aug  6 14:56 F3File.txt
-rw-r--r-- 1 root root 49 Aug  6 15:31 new_file.txt
-rw-r--r-- 1 root root 76 Aug  6 15:31 newfile.txt.gz
-rw-r--r-- 1 root root 15 Aug  6 15:31 source.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# bzip2 F1/F2/F3/new_file.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# ls -ltr F1/F2/F3/
total 16
-rw-r--r-- 1 root root 22 Aug  6 14:56 F3File.txt
-rw-r--r-- 1 root root 85 Aug  6 15:31 new_file.txt.bz2
-rw-r--r-- 1 root root 76 Aug  6 15:31 newfile.txt.gz
-rw-r--r-- 1 root root 15 Aug  6 15:31 source.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#

```

xz: High compression utility.

- Example: `xz file.txt` compresses `file.txt`.
- Description: Provides the best compression ratio.

```
root@techies:/home/nbkx786# ls -ltr *.txt
-rw-rw-r-- 1 nbkx786 nbkx786 561 Aug  6 15:51 new_file.txt
-rw-r--r-- 1 root    root    561 Aug  6 15:52 newfile.txt
-rw-r--r-- 1 root    root    561 Aug  6 15:52 source.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# xz new_file.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# ls -ltr *.txt
-rw-r--r-- 1 root root 561 Aug  6 15:52 newfile.txt
-rw-r--r-- 1 root root 561 Aug  6 15:52 source.txt
root@techies:/home/nbkx786# ls -ltr
total 72
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Templates
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Desktop
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Public
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Documents
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Pictures
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Music
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Videos
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Downloads
lrwxrwxrwx 1 nbkx786 nbkx786    6 Aug  4 15:20 d2 -> d1/d2/
drwxrwxr-x 3 nbkx786 nbkx786 4096 Aug  4 15:22 d1
drwxrwxr-x 3 nbkx786 nbkx786 4096 Aug  4 15:27 m1
drwx----- 7 nbkx786 nbkx786 4096 Aug  6 12:00 snap
lrwxrwxrwx 1 root    root      10 Aug  6 14:45 source_link -> source.txt
drwxr-xr-x 3 root    root      4096 Aug  6 14:50 F1
lrwxrwxrwx 1 root    root       8 Aug  6 14:50 f3_dir -> F1/F2/F3
lrwxrwxrwx 1 root    root      19 Aug  6 14:55 f3file_link -> F1/F2/F3/F3File.txt
-rw-r--r-- 1 root    root    10240 Aug  6 15:32 archive.tar
-rw-rw-r-- 1 nbkx786 nbkx786   128 Aug  6 15:51 new_file.txt.xz
-rw-r--r-- 1 root    root    561 Aug  6 15:52 newfile.txt
-rw-r--r-- 1 root    root    561 Aug  6 15:52 source.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
```

zip: Used for packaging and compressing (archiving) files.

- Example: `zip archive.zip file1 file2` creates an archive with `file1` and `file2`.
- Description: Commonly used for creating `.zip` files.

```

-rw-r--r-- 1 root    root      10240 Aug  6 15:32 archive.tar
-rw-rw-r-- 1 nbkx786 nbkx786    128 Aug  6 15:51 new_file.txt.xz
-rw-r--r-- 1 root    root        561 Aug  6 15:52 newfile.txt
-rw-r--r-- 1 root    root        561 Aug  6 15:52 source.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# zip testarchive.zip newfile.txt source.txt
  adding: newfile.txt (deflated 91%)
  adding: source.txt (deflated 91%)
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# ls -ltr
total 76
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Templates
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Videos
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Music
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Pictures
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Documents
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Public
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Downloads
drwxr-xr-x 2 nbkx786 nbkx786 4096 Aug  3 15:32 Desktop
lrwxrwxrwx 1 nbkx786 nbkx786    6 Aug  4 15:20 d2 -> d1/d2/
drwxrwxr-x 3 nbkx786 nbkx786 4096 Aug  4 15:22 d1
drwxrwxr-x 3 nbkx786 nbkx786 4096 Aug  4 15:27 m1
drwx----- 7 nbkx786 nbkx786 4096 Aug  6 12:00 snap
lrwxrwxrwx 1 root    root       10 Aug  6 14:45 source_link -> source.txt
drwxr-xr-x 3 root    root      4096 Aug  6 14:50 F1
lrwxrwxrwx 1 root    root        8 Aug  6 14:50 f3_dir -> F1/F2/F3
lrwxrwxrwx 1 root    root       19 Aug  6 14:55 f3file_link -> F1/F2/F3/F3File.txt
-rw-r--r-- 1 root    root      10240 Aug  6 15:32 archive.tar
-rw-rw-r-- 1 nbkx786 nbkx786    128 Aug  6 15:51 new_file.txt.xz
-rw-r--r-- 1 root    root        561 Aug  6 15:52 newfile.txt
-rw-r--r-- 1 root    root        561 Aug  6 15:52 source.txt
-rw-r--r-- 1 root    root        424 Aug  6 15:55 testarchive.zip
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#

```

file: Determines the type of a file.

- Example: `file image.jpg` displays the type of `image.jpg`.
- Description: Useful for identifying file formats.

```

root@techies:/home/nbkx786#
root@techies:/home/nbkx786# file newfile.txt
newfile.txt: ASCII text
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# file source_link
source_link: symbolic link to source.txt
root@techies:/home/nbkx786#
root@techies:/home/nbkx786# file f3_dir
f3_dir: symbolic link to F1/F2/F3
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#

```

mount: Attaches a filesystem to a specified directory.

- Example: `mount /dev/sdb1 /mnt` mounts the device `sdb1` to `/mnt`.
- Description: Used to access filesystems on different storage devices.

df: Displays disk space usage of all mounted filesystems.

- Example: `df -h` shows disk space in human-readable format.

- Description: Useful for monitoring disk space usage.

```

root@techies:/home/nbkx786# df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           1.1G  2.9M  1.1G   1% /run
/dev/sda2       20G   11G   8.5G  55% /
tmpfs           2.7G   0  2.7G   0% /dev/shm
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           2.7G   8.0K  2.7G   1% /tmp
none            1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service
tmpfs           537M   80K  536M   1% /run/user/1000
/dev/sr1        6.1G   6.1G   0 100% /run/media/nbkx786/Ubuntu 26.04 amd64
/dev/sr0        108M  108M   0 100% /run/media/nbkx786/CDROM
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#

```

findmnt: Lists all mounted filesystems.

- Example: `findmnt -t ext4` lists all ext4 type mounts.
- Description: Displays a tree-like structure of mounts.

```

root@techies:/home/nbkx786# findmnt
TARGET          SOURCE          FSTYPE          OPTIONS
--
/run            /dev/sda2       ext4            rw,relatime
/run/credentials/systemd-journald.service tmpfs          rw,nosuid,nodev,size=1097856k,nr_inodes=819200,mode=755,inode64
/run/credentials/systemd-resolved.service none           tmpfs          ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=755,inode64
/run/media/nbkx786/Ubuntu 26.04 amd64 none           tmpfs          ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=755,inode64
/run/napd/ns     tmpfs[/snapd/ns] iso9660        ro,nosuid,nodev,relatime,nojoliet,check=s,map=n,blocksize=2048,uid=1000,gid=1000,mode=755,umask=0077
/run/napd/ns/snapd-desktop-integration.mnt nsfs[mnt:[4026532587]] nsfs          rw
/run/napd/ns/promoting-client.mnt nsfs[mnt:[4026533052]] nsfs          rw
/run/napd/ns/firmware-updater.mnt nsfs[mnt:[4026532589]] nsfs          rw
/run/media/nbkx786/CDROM /dev/sr0        iso9660        ro,nosuid,nodev,relatime,nojoliet,check=s,map=n,blocksize=2048,uid=1000,gid=1000,mode=755,umask=0077
/run/user/1000   tmpfs          tmpfs          rw,nosuid,nodev,relatime,size=548924k,nr_inodes=137231,mode=700,uid=1000,gid=1000,mode=755,umask=0077
/run/user/1000/doc gvfsd-fuse      fuse.gvfsd-fuse rw,nosuid,nodev,relatime,user_id=1000,group_id=1000
/run/user/1000/gvfs gvfsd-fuse      fuse.gvfsd-fuse rw,nosuid,nodev,relatime,user_id=1000,group_id=1000
/tmp            tmpfs          tmpfs          rw,nosuid,nodev,size=2744636k,nr_inodes=1048576,inode64,usrquota
/dev            devtmpfs       devtmpfs       rw,nosuid,nodev,relatime
/dev/nqueue     nqueue         nqueue         rw,nosuid,nodev,noexec,relatime
/dev/hugepages  hugetlbfs     hugetlbfs      rw,nosuid,nodev,relatime,pagesize=2M
/dev/shm        tmpfs          tmpfs          rw,nosuid,nodev,inode64,usrquota
/dev/pts        devpts         devpts         rw,nosuid,noexec,relatime,gid=5,mode=600,ptmxmode=000
/snap/bare/5    squashfs       squashfs       ro,nodev,relatime,errors=continue,threads=single
/sys            sysfs          sysfs          rw,nosuid,nodev,noexec,relatime
/sys/kernel/debug debugfs        debugfs        rw,nosuid,nodev,noexec,relatime
/sys/kernel/tracing tracefs        tracefs        rw,nosuid,nodev,noexec,relatime
/sys/fs/fuse/connections fusectl        fusectl        rw,nosuid,nodev,noexec,relatime
/sys/kernel/security securityfs     securityfs     rw,nosuid,nodev,noexec,relatime
/sys/fs/cgroup  cgroup2       cgroup2       rw,nosuid,nodev,noexec,relatime,nsdelegate,memory_recursiveprot,memory_hu
/sys/fs/pstore  pstore        pstore        rw,nosuid,nodev,noexec,relatime
/sys/fs/bpf     bpf           bpf           rw,nosuid,nodev,noexec,relatime,mode=700
/sys/kernel/config configfs       configfs       rw,nosuid,nodev,noexec,relatime

```

lsblk: Lists information about block devices.

- Example: `lsblk` lists all block devices.
- Description: Useful for identifying storage devices and their partitions.

```
root@techies:/home/nbkx786# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
fd0          2:0    1    4K  0 disk
loop0        7:0     0    4K  1 loop /snap/bare/5
loop1        7:1     0  66.8M 1 loop /snap/core24/1587
loop2        7:2     0  66.8M 1 loop /snap/core24/1643
loop3        7:3     0  19.6M 1 loop /snap/desktop-security-center/150
loop4        7:4     0   20M  1 loop /snap/desktop-security-center/151
loop5        7:5     0 273.7M 1 loop /snap/firefox/8107
loop6        7:6     0 606.1M 1 loop /snap/gnome-46-2404/153
loop7        7:7     0  16.5M 1 loop /snap/firmware-updater/226
loop8        7:8     0  18.8M 1 loop /snap/prompting-client/204
loop9        7:9     0  15.7M 1 loop /snap/snap-store/1367
loop10       7:10    0  50.1M 1 loop /snap/snapd/27591
loop11       7:11    0   395M 1 loop /snap/mesa-2404/1165
loop12       7:12    0  91.7M 1 loop /snap/gtk-common-themes/1535
loop13       7:13    0   580K 1 loop /snap/snapd-desktop-integration/361
loop14       7:14    0  49.3M 1 loop /snap/snapd/26865
loop15       7:15    0  18.8M 1 loop /snap/prompting-client/222
loop16       7:16    0   828K 1 loop /snap/snapd-desktop-integration/391
sda          8:0     0   20G  0 disk
├─sda1       8:1     0    1M  0 part
└─sda2       8:2     0   20G  0 part /
sr0         11:0     1 107.9M 0 rom  /run/media/nbkx786/CDROM
sr1         11:1     1   6.1G  0 rom  /run/media/nbkx786/Ubuntu 26.04 amd64
root@techies:/home/nbkx786#
root@techies:/home/nbkx786#
```